

LECTURE 3

REQUIREMENTS

GATHERING

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READINGS

Beginning Software Engineering chapter 4

<https://onlinelibrary-wiley-com.libproxy1.nus.edu.sg/doi/pdf/10.1002/9781119209515.ch4>

LEARNING OBJECTIVES

At the end of this lecture, you should:

- Recap the types of requirements and their properties
- Apply requirements gathering techniques to identify use cases
- Understand How to draw Use Case Diagrams

RECAP FROM PRE-CLASS VIDEO

In the pre-class video, we learned about:

- What Requirements are and their importance.
- The 3 Types of Requirements:
 - Functional (The "What")
 - Nonfunctional (The "How Well")
 - Implementation (The "Transition")
- 3 Key Gathering Approaches: Interviewing, Analyzing Processes, and Studying Systems.

Today, we'll focus on applying these concepts.

PROPERTIES OF REQUIREMENTS

Clear

Unambiguous

Comprehensive / Exhaustive

Consistent

Prioritized

Verifiable

UNDERSTANDING BUSINESS PROCESS FLOW

Often, we can analyze business process flow using **events**

- Not all events in a business process is relevant
- Only those that should be **remembered by the system** or those that should **trigger processing in the system** are important

3 TYPES OF EVENTS



External Events

- Occur outside the system
- **Usually caused by external agent**
e.g. customer

Temporal Events

- Occurs when system reaches a point (deadline) in **time**

State Events

- Events that occur when something happen inside the system that **trigger the need for processing**

EXTERNAL EVENTS

Examples:

- Customer wants to login
- Customer want to check account details
- Customer want to show the list of transaction
- Customer want to search for items
- Etc

TEMPORAL EVENTS

Examples:

- Time to produce monthly report
- Time to email bill PDF
- Time to send scheduled advertisements
- Etc

STATE EVENTS

Examples (triggered when certain conditions are satisfied):

- *(when temperature increases beyond the threshold)* Send a warning notification
- Etc

USE CASES

We then document these requirements as **use cases** (i.e. system functionality)

Example:

- Requirement: **Customer want to show the list of transactions**
- Use Case: **Get list of transactions of customer**
- Requirement: **Customer want to buy item**
- Use Case: **Create new order**

USE CASE DIAGRAMS



Introduction to
Requirements

Types of
Requirements

Requirements
Gathering
Approaches

Use Case
Diagrams

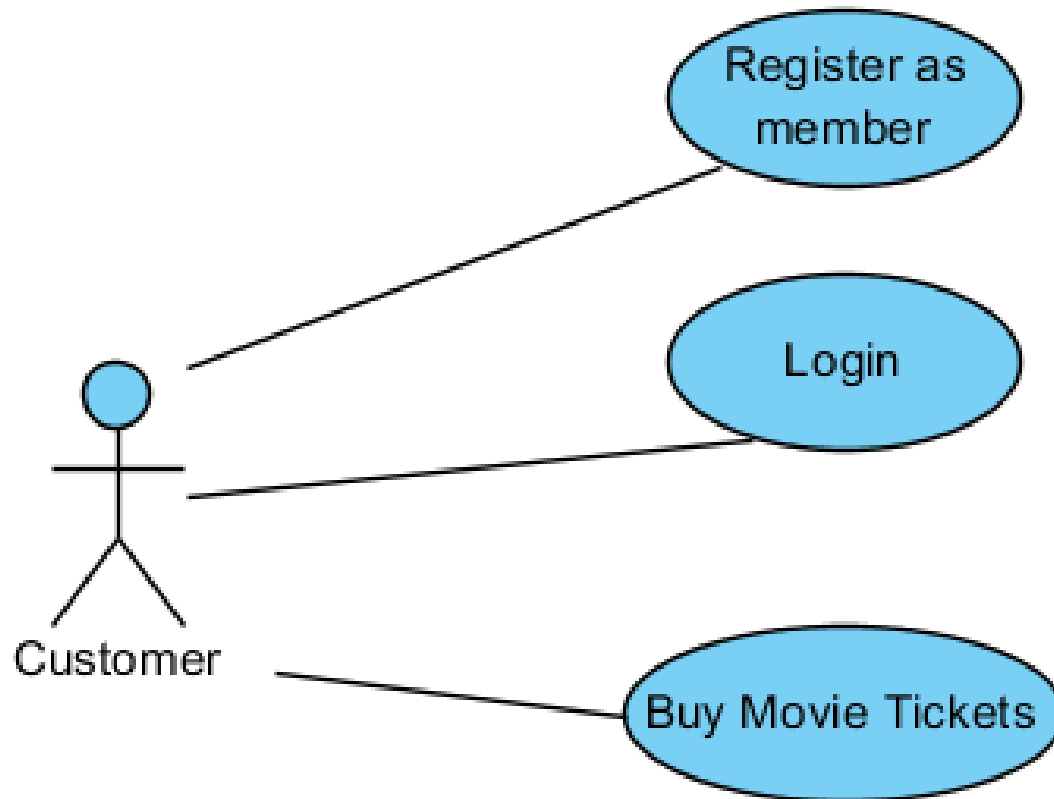
USE CASE DIAGRAMS

Use case diagrams is a **modeling technique** for documenting the **use cases**

Diagram consists of 3 main elements

- Actors
- Use cases
- Association between actors to use cases

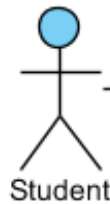
AN EXAMPLE OF A USE CASE DIAGRAM



IDENTIFY ACTORS

Actors

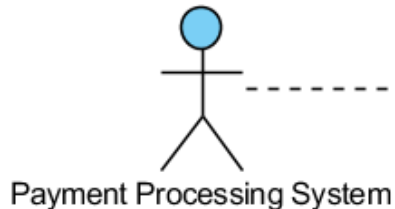
- Can be human users



Example of a human actor

(This is a comment)

- Or other systems that interact with this IT system



Example of an external computer system actor

(Also drawn using the stick figure)

GENERALIZE ACTORS

Questions to ask

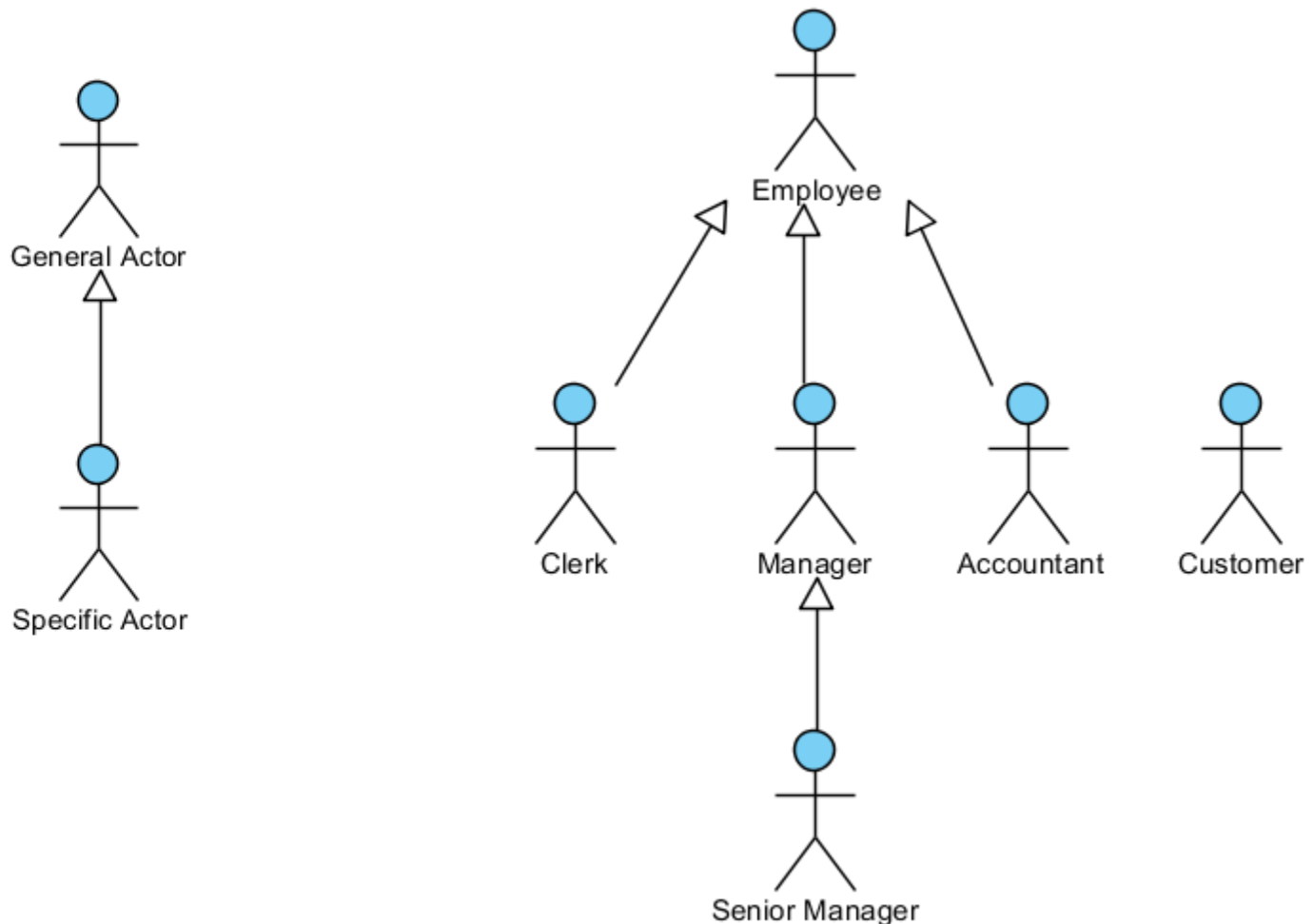
- Who are the different types of users?
- Employees, Managers, Accountants, etc

Generalize them

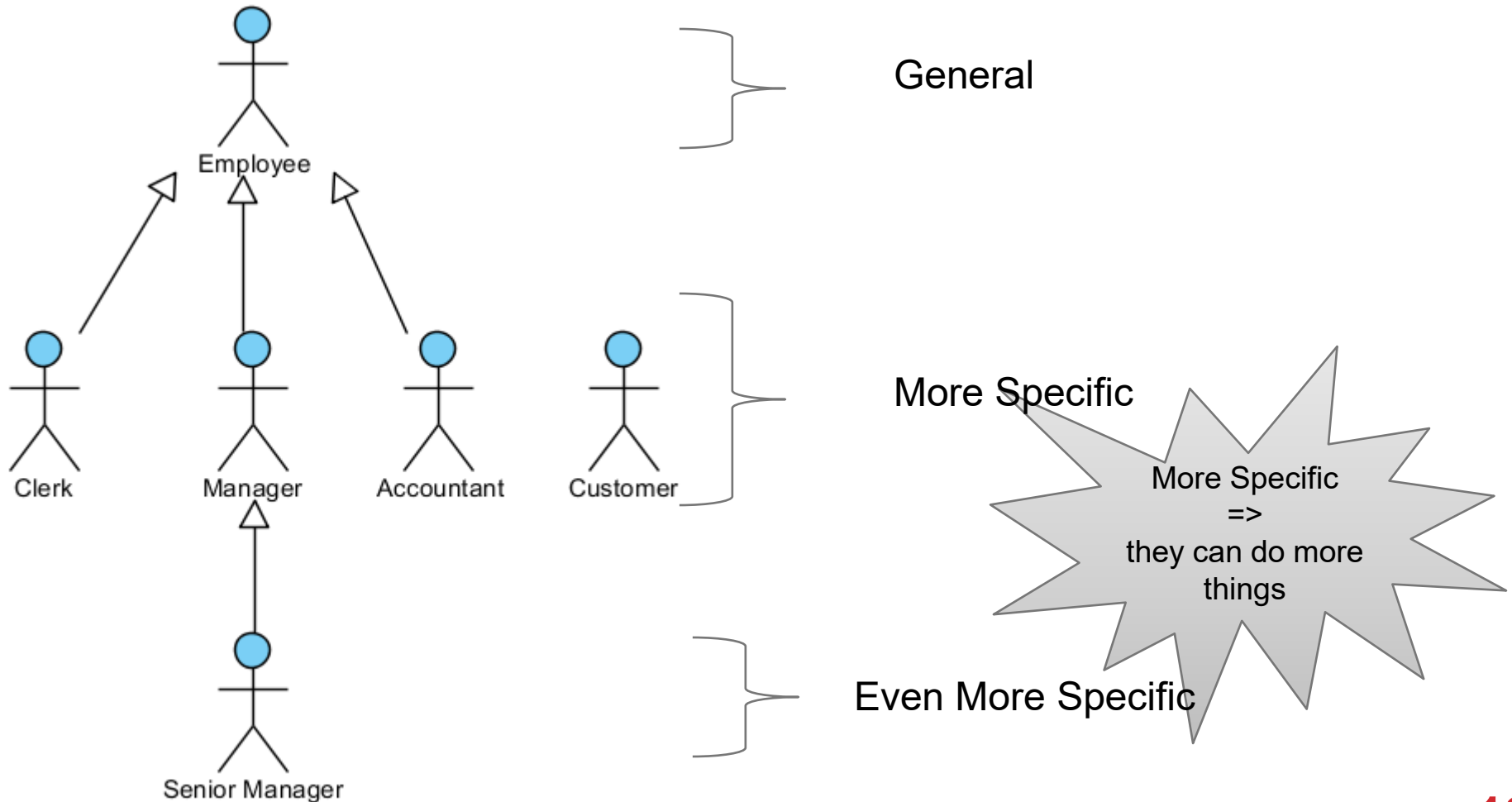
- i.e. find relations amongst these different types
- e.g. an undergrad is a more specific kind of student

ROLE MAP

Recall: Superclass and subclass in OO programming



ROLE MAP



GENERALIZATION AND SPECIALIZATION

Specialized actors inherit the ability to do all the things that a generalized actor can do

Think Object-Oriented programming

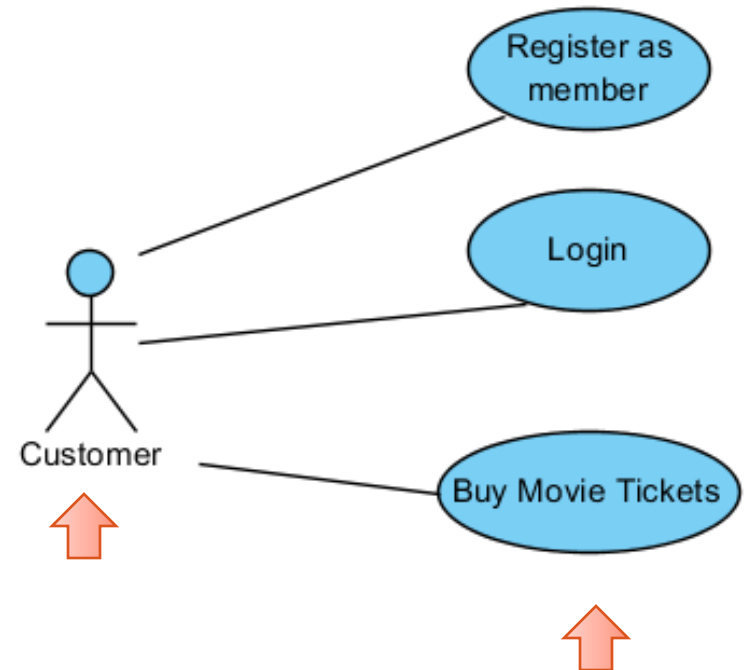
Clerk extends (inherits) Employee

- Employee = Generalized actor
- Clerk = Specialized actor

USE CASE MODELING ELEMENTS

Actor: An actor who initiates a use-case interaction.

Use case: A user task (indicated as an oval)



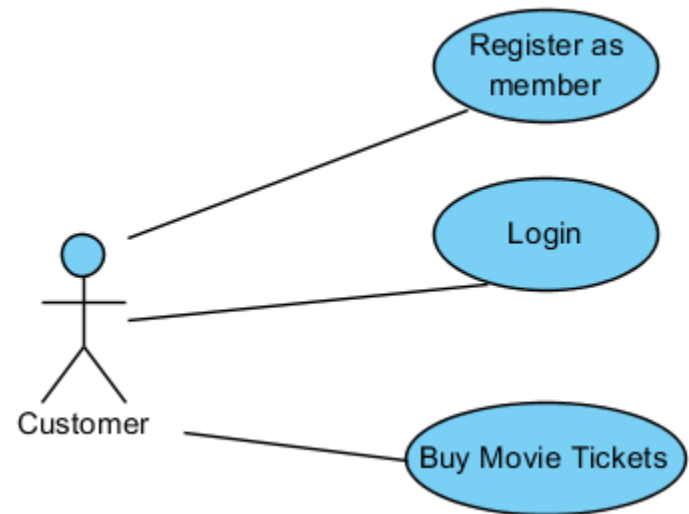
Can draw with either a line or arrow pointing to the use case

IDENTIFY USE CASES

A use case is an **interaction** between an actor and a computer system

Each use case is named using a verb

E.g. “Register as user”, “Login”,



GUIDE FOR IDENTIFYING USE CASES

1. Need to identify all the actors of this system

- Note that actors need not be just human users, they can be **other systems** that interact with this system

2. A use case must be a system function/interaction

- E.g. clerk calling a customer is NOT a use case - it's not a system function

3. A use case should fulfil a business objective

- E.g.
Actor: Student
Use case: View list of enrolled courses

GUIDE FOR IDENTIFYING USE CASES

4. While a use case is a system function/interaction, need to know how to **differentiate between a use case vs a step of a use case**

- Note that a use case is a high-level system function
- We will derive the individual steps of a use case in future phases (not now)
- Do NOT try to list out the individual steps of a use case
- E.g
Use case: Update user profile
Possible steps of the “Update user profile” use case:
enter name, enter date of birth, enter email, etc
- Keep in mind that a use case must fulfil a business objective
- Ultimately whether a system interaction is a use case, or a step depends on the design of the system (will talk more about this later on)

GUIDE FOR IDENTIFYING USE CASES

5. **Use cases are not just a function in code. It should be seen from a point of view of an actor**
 - E.g. “check password during login”, “send email confirmation”, “add to database”, etc are not use cases. They are what the system/code does
6. **Use cases are seen from the perspective of a user rather than from the perspective of the system**
 - i.e. “get a list of chat messages” rather than “receive a chat message”
7. **A use case diagram should not have this system itself as an actor**
 - Usually, we omit time event in the use case diagram
 - But if you really want to draw, you can create an “actor” called TIME

QUESTION TIME



1. Why do we gather requirements using a use case diagram rather than just come up with a list of system functions?

A man with dark hair and glasses is holding a white rectangular sign with both hands. He is looking directly at the camera with a slight smile. The sign has the text "It's your turn..." written on it in a black serif font. The background is a plain, light-colored wall.

It's your turn...

GROUP WORK (15 MINS)

In groups (3 – 4 members):

Come up with the Use Case Diagram for the Canvas system (focusing just on the Discussion Forum Feature only)

Normally, this process will take quite some time, but here we will just stop after 15 mins

UPLOAD your final work to Canvas
(Assignments > L3 Use Case Diagram Exercise)

DEBRIEF



SUMMARY

Properties of Requirements

**Types of Requirements : Functional, Nonfunctional,
Implementation Requirements**

Requirements Gathering Approaches

Role Map and Use Case Diagrams

WHAT'S NEXT?

Requirements Analysis